# ENGO 625 – Advanced GNSS Theory and Applications (F2025)

Lab #4 – Carrier Phase Float Solution

(Instructor: Kyle O'Keefe)

## Due Date

Monday, 1 December 2025 at 4:30 PM. Please submit your written report and source code.

## Objectives

1. To become familiar with phase processing.
2. To understand the abilities of carrier phase differential processing using a float solution in L1 only mode.
3. To improve programming skills with regards to Geomatics Engineering (C/C++, MATLAB or Python, please).

## Data Description

Use the data you were provided for Labs 1-3.

## Tasks

1. Implement the phase rate method for cycle slip detection on L1
   a. Plot any cycle slips as a function of time.
   b. Identify a data set of approximately 20-30 minutes with the following characteristics:
      - No cycle slips.
      - No blunders (refer to Labs 1, 2 and 3).
      - No satellites dropping in and out (so that you can choose one and only one base satellite).
      - Adequate geometry and number of satellites for ambiguity resolution.

      It will likely be necessary to remove an entire sequence of observations from one or more satellites in order to meet the above criteria. Justify the data set you have chosen.
2. From the data set you have selected, use double-differenced L1 phase observations and pseudoranges to perform either a sequential least-squares (SLSQ) solution or a Kalman filter (KF) solution – the choice is yours. If you select a KF, justify the dynamics model that you use. You may also implement this as a between receiver single difference solution, but then special care is required for the receiver clock dynamics model. It needs potentially a large amount of process noise because it is not a "random constant"

3. Estimate the remote station position and the **float** ambiguities.
   a. Using the true position given to you, determine the position errors in the East-North- Up (ENU) frame.
   b. Plot the time series of each of these error components (East, North, Up) **on separate plots** (use the subplot function in Matlab or something similar in Python). On the same graphs, plot the time series of the estimated standard deviations. You must plot both the standard deviation (which is always positive) and the standard deviation multiplied by negative one (which will then be always negative) in order to get the estimated accuracy "envelope".

c. Discuss the differences between the true and estimated accuracy. Comment on which assumptions you made may influence any discrepancies between the two. Justify your assumptions if necessary.
d. Comment on the differences in accuracy between this solution and the differential pseudorange-only solution from Lab 3.

## Report Format

Each report must be professionally written. Be concise but thorough. Explanations should accompany all plots. Assumptions must be clearly documented and justified.

## Grading Criteria (7% of Final Grade)

Technical Accuracy (5/7)

- Correct solution/plots
- Correct analysis of results
- Correct justification and assumptions
- Insight into results
- Graphs correctly labeled (each axis with out a unit or title will be -1 point)

Source Code (1/7)

- Methodical and logical procession of algorithms
- Commented source code
- If you share software with other classmates, you must acknowledge their contribution both in the comments and in the text of the report.

Report Quality, Neatness and Readability (1/7)

- Neat, clear and effective use scales on plots
- Logical organization of lab

## References

ENGO 625 Course Lecture Notes - Advanced GNSS Theory and Applications
ENGO 421 Course Lecture Notes (might be useful for coordinate systems)
ENGO 363 Course Lecture Notes or Text (for a detailed explanation of Least Squares)